DATA REPORT · MARCH 2026

THE STATE OF AI-DRIVEN SEO IN NUMBERS.

We analyzed 341,701 content pieces across 57 brands, using direct portfolio comparisons as the main evidence base and an exploratory ML appendix on 61,790 active content records. The goal is a public-safe study that a reader can act on without FlyRank tooling.

flyrank.com

About This Study

This paper examines 341,701 content pieces across 57 brands, spanning 469,879,632 search impressions, 1,514,819 clicks,1,635,404 sessions, and 17,344 AI sessions.

Rather than reproducing dashboard screenshots or internal table views, we tested portfolio-level questions about content age, freshness, position, traffic mix, depth, and AI visibility. Direct aggregate comparisons lead the paper. Machine-learning pages appear later as exploratory appendix material.

The goal is practical clarity, not maximum claim volume. We keep the strongest, most reproducible patterns and demote or remove weaker internal constructs when they are not suitable for a public paper.

Each finding ends with manual next steps a reader can execute using standard analytics, search, and content-review workflows.

Thesis

"The most useful SEO decisions in this portfolio came from simple, validated comparisons: refresh strong content before it decays, protect page-one assets, and interpret composite scoring through raw search performance rather than in place of it."

341,701

content pieces analyzed across 57 brands

90-DAY IMPRESSIONS	90-DAY CLICKS	SESSIONS TRACKED
469.9M	**1,514,819**	**1.6M**
Google Search Console	Organic search	GA4 data

AI SESSIONS	AI SHARE	AVG HEALTH SCORE
17,344	**1.06%**	**22.3**
Known AI referrals	Of tracked sessions	FlyRank composite context

30-DAY TREND VS PREVIOUS 30 DAYS

+70.6%	+51.9%	+34.2%
Impressions	Clicks	AI Sessions

Chart read: These cards compare the last 30 days with the 30 days before that. Read them as short-term momentum signals for the portfolio, not as a substitute for the longer 90-day comparisons used in the main findings.

The portfolio spans newly published pages, established evergreen assets, and underperforming long-tail pages. That range is what makes the lifecycle, freshness, and visibility patterns in this paper useful rather than anecdotal. Scope cards use the local 90-day snapshot for performance metrics, while inventory fields such as content count, age, and word count remain metadata for the full cached portfolio.

Study Scope & Evidence Standard.

Verified totals, reporting windows, and interpretation rules for the rest of the paper.

This study evaluates the all-portfolio export first. The current paper covers 341,701 content pieces across 57 brands, with 469,879,632 impressions, 1,514,819 clicks,1,635,404 sessions, and 17,344 AI sessions.

Reporting windows: 90-day performance window (rolling), 30-day trend comparison, local historical monthly series, and local active-content feature-vector cuts where noted.

VERIFIED TOTAL	VALUE
Content	341,701
Brands	57
Impressions	469,879,632
Clicks	1,514,819
Sessions	1,635,404
AI Sessions	17,344

Metric Windows

90-day performance window (rolling), 30-day trend comparison, local historical monthly series, and local active-content feature-vector cuts where noted.

Local Snapshot Rule

This paper uses the local cached snapshot and local derived exports. Some extended cuts come from the local full feature vector, which is an active-content subset with impressions_90d > 0 and sessions_90d > 0.

Evidence Standard

Headline findings prioritize direct aggregate comparisons. ML pages are exploratory appendix material and do not override direct portfolio evidence.

Interpretation Rule

External SEO beliefs are treated as hypotheses, not proof. When direct portfolio evidence conflicts with industry narratives, the portfolio result wins.

Understanding the Metrics

The paper leads with raw search performance. Composite and ML metrics are included only when they add context.

Health Score (0-100)

Impressions (30 pts) + position (30 pts) + CTR (20 pts) + scroll depth (20 pts). It is a FlyRank summary metric, not a market-standard outcome metric, so major findings pair it with raw search performance.

Position Tiers

Top 3: positions 1-3 | Page 1: positions 4-10 | Striking Distance: positions 11-20 (highest ROI optimization zone) | Page 3-5: positions 21-50 | Deep: 50+

Age & Freshness

Age: days since content creation. Freshness: days since last content update. These are independent — old content can be freshly updated, and new content can already be stale if never optimized.

AI Traffic

Sessions from known AI referrals such as ChatGPT, Perplexity, Gemini, Copilot, Claude, and Meta AI. Currently 1.1% of tracked sessions (17,344 of 1.6M).

Optimization Flags

Internal FlyRank workflow labels such as Healthy, Fix CTR, Fix Content, and Zombie Page. They are triage cues, not public definitions of content quality.

Trend Direction

Calculated from 30d-vs-prev-30d impression change. Up: >10% growth. Down: >10% decline. Stable: within +/-10%. Flat: insufficient data. New: content created within 30 days.

Finding Tags

CONFIRMED Hypothesis validated by data

REVERSED Data showed the opposite

NUANCED Partially true with caveats

ML Techniques Used

K-Means clustering, Random Forest, logistic regression, PCA, and a shallow decision tree appear only in the exploratory appendix. Minimum sample size is 50 per bucket (30 for cross-correlations), and weighted CTR is used where total clicks and impressions are available.

The Anatomy of Growing Content.

Hypothesis: content trending upward has a different structural profile than content trending downward.

The first comparison in the study is deliberately simple: pages with rising impressions versus pages with falling impressions. In this portfolio, the main differences are depth, age, and visibility. Growing pages tend to be longer, younger, and already positioned slightly better in search.

3180 avg words (growing)	**2311** avg words (declining)	**184d** avg age (growing)	**230d** avg age (declining)

DIRECTION	COUNT	WORDS	AGE	AVG IMP	AVG POS	HEALTH
up	74.8K	3.2K	184d	2.8K	15.9	36.4
down	45.6K	2.3K	230d	2.6K	16.1	32.4

Growing content is 37.6% longer (3.2K vs 2.3K words) and **20% younger** (184 vs 230 days). Interestingly, declining content is not simply "bad content" in the abstract. Many declining pages still carry meaningful impressions, but they are older, thinner, and less likely to retain momentum. The sample sizes here are large (74,187 rising vs 45,272 falling), so the word-count and age gap is directionally robust even though this remains an observational comparison.

What to Do With This

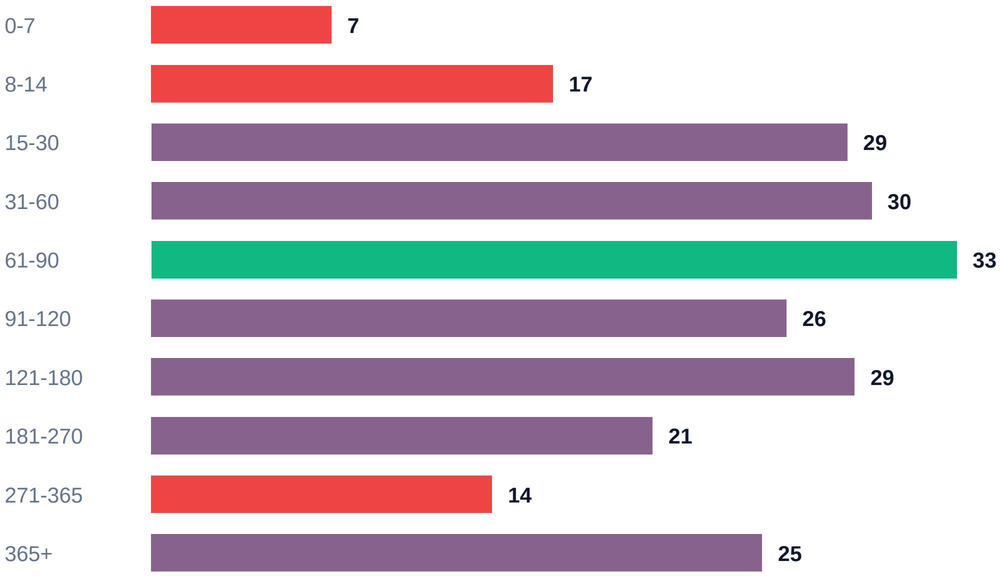
Expand thin pages that already earn impressions
Why: The rising cohort is materially longer than the declining cohort
How: List pages under roughly 2,000 words that still earn meaningful impressions, then add missing subtopics, examples, definitions, and comparison sections instead of padding.
Expected: Improves the odds that an already visible page can keep growing rather than plateauing.
Measure: Track impressions, clicks, and average position 30 and 60 days after the update.

Review aging pages before they drift into decline
Why: The falling cohort is older on average and more likely to have missed a recent refresh cycle
How: Create a quarterly review for pages that are around 6-9 months old, then prioritize the ones that once performed well but have started to flatten.
Expected: Protects pages with existing visibility before the decline becomes expensive to reverse.
Measure: Track refresh coverage and 30-day impression change for reviewed versus unrevised pages.

The Content **Performance Curve.**

Content peaks at 61-90 days, declines after 270 days, and the 365+ rebound is concentrated in older pages that were refreshed.

HEALTH SCORE BY CONTENT AGE (DAYS)



The data reveals a clear lifecycle pattern. Content enters a **growth phase** during its first 90 days as Google discovers, indexes, and begins ranking it. It hits **peak performance at 61-90 days** (health score 33.1).

A **maturation plateau** follows from 91-180 days where performance stabilizes. Then comes the **decay cliff at 271-365 days** (health drops to 14) — the point where content staleness overtakes accumulated authority.

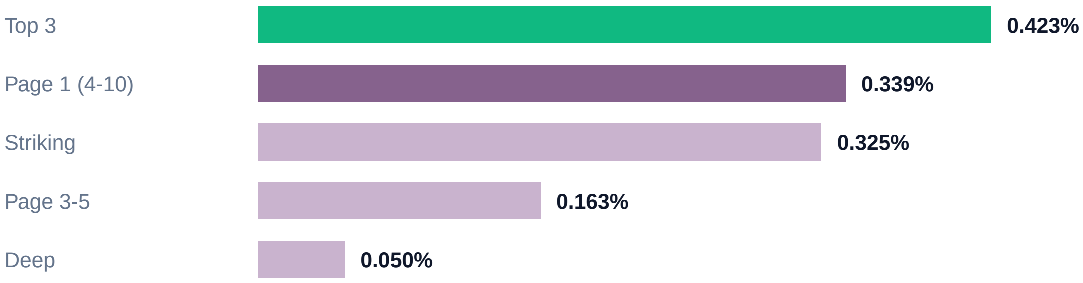
The recovery at 365+ (25.1) is still actionable, but the safe reading is narrower: older pages can recover when they are updated well. This is not evidence that age naturally reverses performance decline on its own.

Create a review cycle before pages hit the 9-12 month decay zone
Why: The steepest drop appears once pages move past the stronger mid-life window
How: Schedule editorial reviews for pages approaching that age range, then refresh the ones with proven demand or strategic importance before they go stale.
Expected: Helps preserve the visibility that older pages have already earned.
Measure: Track refreshed-page coverage and compare impression retention before and after the review window.

Click Capture by Position Tier.

Weighted portfolio CTR declines sharply as visibility moves away from the top of the results.

WEIGHTED CTR BY POSITION TIER



0.423%
weighted CTR in top 3

0.339%
weighted CTR on page 1

88%
drop from top 3 to deep

These are **portfolio-level weighted CTRs** computed from total clicks divided by total impressions in each position tier. They are not generic Google CTR benchmarks, and they replace the older per-row average that produced impossible values above 100%.

The pattern still matters: once content leaves the best positions, click capture compresses quickly. That makes page-one refinement a more reliable source of incremental clicks than spreading effort evenly across pages with little existing visibility.

Refine search snippets on pages that already rank on page 1
Why: Small ranking and snippet gains on already-visible pages usually convert into click lift faster than rebuilding weak pages from scratch
How: Review pages ranking on page 1, tighten titles and descriptions around clear intent, and test more specific promise statements instead of generic headings.
Expected: Improves click capture on visibility you already own.
Measure: Track weighted CTR, clicks, and average position for the revised group over the next 30 days.

Support positions 11-20 with targeted relevance improvements
Why: Pages just outside page 1 already show search demand and topical fit
How: Refresh the page, improve internal links from stronger related pages, and tighten the opening sections so the page answers the target query faster.
Expected: Increases the chance of moving a borderline page into the higher-click part of the curve.
Measure: Track movement into page 1 and the resulting click lift.

The Freshness Multiplier.

The 31-90 day window is the strongest stable freshness band. The `361+` bucket is visible in the chart, but it is too small and too unstable to treat as a headline multiplier.

GROWTH-TO-DECLINE RATIO BY FRESHNESS WINDOW

The 0-30 day tier is still too early for a stable momentum read. But at **31-90 days, freshness remains the strongest measured growth window** at 7.88:1. Splitting the old stale bucket also improves the story: pages that have been untouched for **181-360 days** are a different population from the **361+ day** tail. In the local active-content sample, `181-360` still reads like a real stale bucket at 3.13:1, while `361+` spikes to 283:1 only because the sample is tiny and there is just 1 declining page in that bucket.

The most dramatic finding: 365+ day content that was refreshed within 30 days shows **3.2x health boost** (from 10.7 to 34.5) and **57x more impressions** (from 71 to 4039). In this portfolio, refresh timing is one of the strongest measured levers available.

3.2x health boost from refresh	**57x** impression boost	**7.88:1** growth ratio at 31-90d

Chart read: Read 31-90 as the strongest stable freshness signal. The 361+ bar is present because it exists in the local sample, but its 283:1 ratio is unstable: 283 growing pages versus only 1 declining.

Run a recurring refresh program for mature pages

Why: Refreshing mature pages produces 3.2x health and 57x impressions in this dataset

How: Review older pages with proven historical visibility, refresh outdated facts and examples, add missing subtopics, and resubmit the strongest updates for recrawl.

Expected: Lifts mature pages from roughly 10.7 health to 34.5 and materially improves impression potential.

Measure: Track impressions, clicks, and average position on refreshed mature pages versus comparable untouched pages.

Engagement and Visibility Move Together.

High scroll + high engagement = +11.2 health points. Visibility consistency compounds the effect.

SCROLL × ENGAGEMENT	COUNT	HEALTH	AVG IMP	AI %	POS
high_scroll × high	1.0K	50.7	629	5.75	15.9
high_scroll × mid	3.6K	50.3	5.3K	1.78	14.3
low_scroll × mid	10.7K	49.4	11.8K	1.34	14.2
high_scroll × low	13.6K	46.7	1.1K	1.43	15.2
low_scroll × low	32.9K	39.5	1.7K	1.53	14.2

HEALTH BY VISIBILITY CONSISTENCY (DAYS VISIBLE IN 90)

In this portfolio, stronger engagement patterns and steadier visibility appear together. Content visible 80+ days scores 46.8 health. Sporadic content scores 28.4. **The gap is 18.4 points — nearly a full tier difference.**

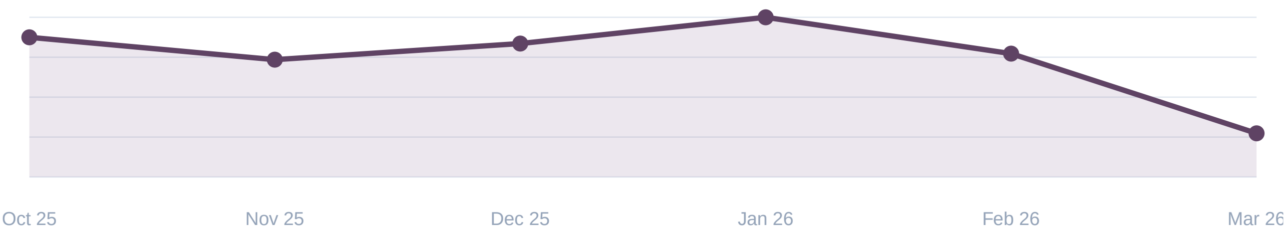
Improve scanability on pages that already earn consistent visibility
Why: The strongest engagement bucket materially outperforms the weakest one in this dataset
How: Use clearer sectioning, helpful jump links, tighter intros, and supporting visuals where they make the page easier to navigate and understand.
Expected: Supports the engagement patterns associated with stronger visibility.
Measure: Track scroll depth, clicks, and average position changes on the revised group.

AI Traffic: A Different Signal.

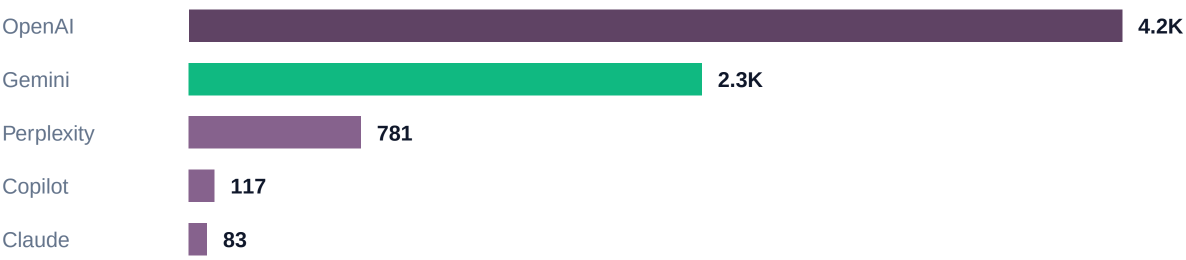
AI referrals are still small in portfolio terms, but the pages that attract them behave differently from classic organic winners.

AI BUCKET	COUNT	HEALTH	AVG IMP	AVG POS
high_ai	873	47.4	24.9K	19.8
some_ai	3.8K	46	10.4K	17
no_ai	57.1K	43.3	2.7K	14.2

AI TRAFFIC SHARE (%) — MONTHLY



AI REFERRAL PROVIDERS (LOCAL LAST-30D BREAKDOWN)



1.06%
AI share of portfolio sessions

7.59%
AI page rate in active sample

1.8x
OpenAI vs Gemini (30d)

The AI bucket behaves differently from classic organic winners. In this dataset, the high-AI group averages about 9x more impressions, yet its average Google position is weaker than the no-AI group. That suggests AI-referral visibility is not simply a mirror of Google rank.

The proportional caveat stays important: portfolio AI traffic is only 1.06% of tracked sessions (17.3K of 1.6M). Inside the local active-content sample, the highest AI page rates show up in commercial intent and the 5K+ word-count bucket. The safe conclusion is narrower: AI-referral visibility is real, growing, and behaviorally different from classic organic traffic.

Treat AI referrals as a separate visibility layer, not a portfolio headline on their own

Why: AI traffic is still a small base, but the pages that attract it look structurally different from classic organic winners

How: Review high-impression reference pages, tighten factual clarity, surface named evidence early, and monitor AI referrals separately from search clicks.

Expected: Improves AI-referral readiness without overstating the size of the channel.

Measure: Track AI session share, provider mix, and page-level AI referral incidence alongside standard search metrics.

The Winning **Combinations.**

Intent, competition, and depth matter most when they work together rather than in isolation.

AVERAGE HEALTH BY INTENT × COMPETITION LEVEL

Chart read: Each bar is one intent-and-competition pairing. Longer bars mean that combination tends to produce stronger pages in this portfolio, so start by copying the top configurations before experimenting with harder ones.

PAGE-1 CONTENT BY WORD COUNT TIER

Chart read: This second chart only looks at pages already on page 1. Use it to judge how much depth winning pages tend to need, not as a rule to make every page longer.

The cleanest combination in this portfolio is transactional intent with low competition. That bucket leads the intent table on both health (30) and impressions (3.0K), while informational low-competition pages average 22.1 health and 1.5K impressions.

On page 1, longer content is not automatically better, but the strongest page-one word-count bucket in this export is **3500+ words**: 44.8 health and 5.0K average impressions versus 40.1 health and 955 average impressions for the weakest page-one word-count bucket. Depth appears most useful when a topic genuinely needs it and the page is already competing for meaningful demand.

> **Reserve a larger share of the calendar for lower-competition commercial or transactional topics**
> Why: Those topics outperform the portfolio average in both visibility and FlyRank health context
> How: Map topics where the reader is already comparing options, then target the versions where the current result pages are beatable with a stronger page rather than a bigger budget.
> **Expected: Improves the odds of reaching page 1 faster and capturing more useful demand.**
> Measure: Track impressions, clicks, and average position by intent class over the next publishing cycle.

The Age-Freshness Matrix.

Content performance is determined by both age and freshness, and the stale side is clearer once `181+` is split into `181-360` and `361+`.

HEALTH SCORE HEATMAP: AGE TIER × FRESHNESS TIER

	0-30	31-90	91-180	181-360	361+
0-14	40.3				
15-30	42.3				
31-90	44.1	44.3			
91-180	41.6	38.5	45.2		
181-365	45.5	30.6	49.7	38.8	
365+	44.6			31.6	46.4

Four quadrants:

1. **Young + Fresh** (31-90d age, 0-30d fresh): Health 44.12 — the growth engine.
2. **Old + Refreshed** (365+, 0-30d fresh): Health 44.62 — proof refresh works.
3. **Mid + Stale** (181-365d, 181-360 fresh): Health 38.75 — the main decay zone.
4. **Old + Long-Unchanged** (365+, 361+): Health 46.36 in a very small active-content survivor sample.

The most powerful insight: **old content that gets refreshed performs nearly as well as new content**. The 365+ refreshed quadrant (44.62) is close to the peak young-fresh quadrant (44.12). The small `365+ × 361+` cell should not be used as a headline decay proof point because the active-content subset introduces strong survivor bias there.

Captured Traffic Value.

The defensible proxy here is `clicks × CPC`, not `impressions × CPC`.

CLICK-EQUIVALENT VALUE BY INTENT

INTENT	PAGES	CLICKS	CPC	VALUE
informational	36.8K	486.9K	0.15	$74.9K
commercial	9.9K	127.3K	0.68	$86.1K
transactional	10.5K	121.7K	0.76	$92.0K
navigational	108	1.6K	0.35	$549

$253.5K
captured click-equivalent value

$73.0M
unsafe impressions × CPC

clicks × CPC
recommended formula

This is a value proxy, not booked revenue. CPC is a cost-per-click benchmark, so the defensible portfolio read is **captured click-equivalent value** from clicks × CPC. Using impressions × CPC inflates the number dramatically because impressions are not billable clicks.

In the local active-content sample, transactional intent leads this captured-value view, followed by commercial intent. That makes value density clearer than raw traffic volume alone.

Use click-equivalent value to rank optimization candidates
Why: It preserves CPC economics without pretending every impression is worth a click
How: Prioritize visible pages where clicks already exist and the underlying CPC is meaningful, especially in transactional and commercial intent buckets.
Expected: Improves prioritization quality when traffic counts alone hide value density.
Measure: Track click lift, weighted CPC, and click-equivalent value before and after optimization.

AI Model **Performance.**

The honest comparison is age-controlled. OpenAI and Gemini cohorts each lead in some windows, so the result is exploratory rather than absolute.

AGE-CONTROLLED COHORT HEALTH: OPENAI VS GEMINI

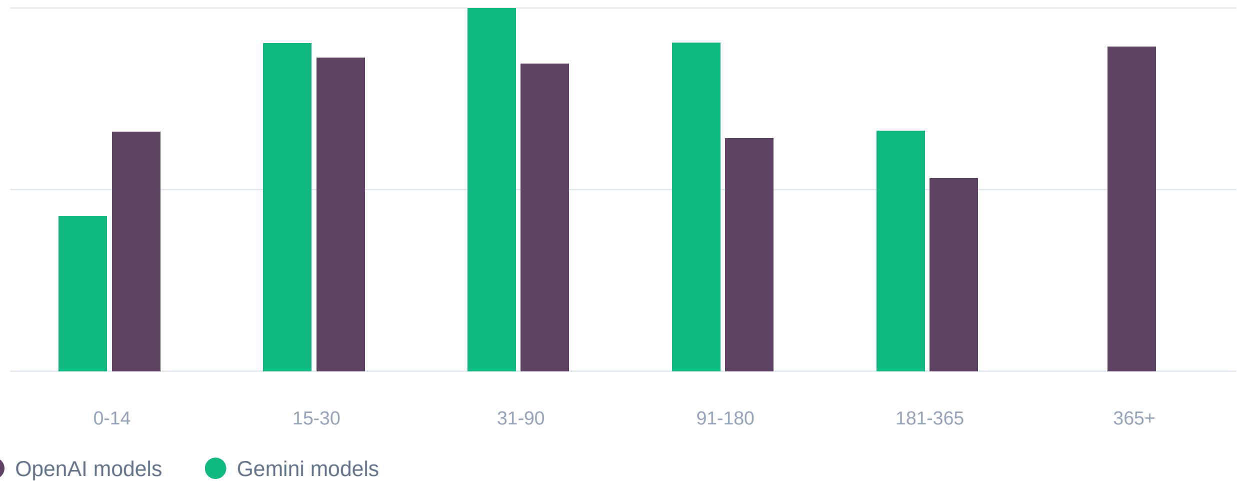


Chart read: Each age tier compares provider families inside the same publication band. Read it as a cohort comparison, not as proof that one provider family wins everywhere.

PROVIDER	COUNT	HEALTH	AVG IMP	AVG POS
OpenAI	145.5K	18.92	689	15.7
Gemini	91.4K	27.13	1.6K	15.6

This section is best read as a **process comparison, not a victory lap for one provider family.**Once age mix is controlled, Gemini leads some cohorts and OpenAI leads others. That is a cleaner view than the older dense table, but it still does not justify a blanket claim that one model family universally wins.

The safer conclusion is that output quality, editing, topic fit, and rollout timing matter more than a simple AI-versus-human framing. This dataset does not support a blanket penalty narrative tied only to AI usage.

Compare content systems within the same age and topic cohorts
Why: The model picture changes once age mix is controlled
How: When you test writing systems or prompts, review them inside the same publication window and topic class rather than comparing a newer batch to an older installed base.
Expected: Produces cleaner decisions about which production workflow actually performs better.
Measure: Track impressions, average position, and health context within matched cohorts at 30, 60, and 90 days.

Portfolio-Level Trends.

How portfolio metrics evolved over 6 months — impressions, clicks, sessions, and AI traffic.

MONTHLY IMPRESSIONS

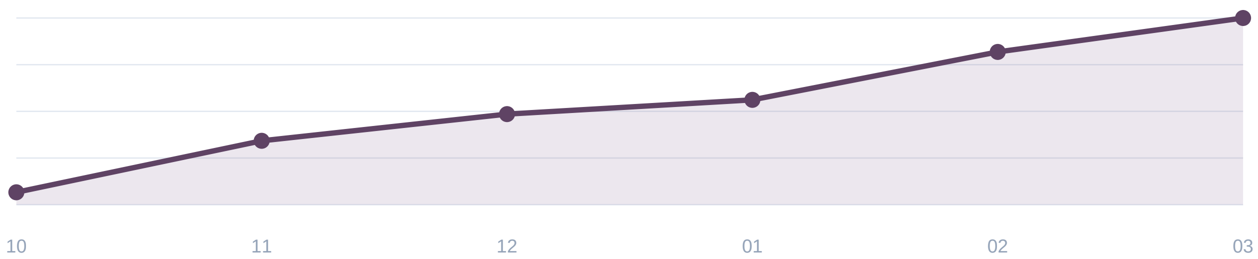


Chart read: Read this as a portfolio trend line, not a page-level benchmark. A rising line means the overall content base is earning more visibility over time, while a flat or falling line signals slowing momentum.

MONTH	IMPRESSIONS	CLICKS	SESSIONS	AI SESSIONS	ACTIVE CONTENT
2025-10	12.5M	75.6K	22.2K	422	17.3K
2025-11	64.4M	271.7K	229.9K	3.7K	93.5K
2025-12	91.3M	297.2K	208.2K	3.8K	98.4K
2026-01	105.7M	375.4K	219.4K	4.8K	110.0K
2026-02	154.0M	508.7K	345.3K	5.8K	156.4K
2026-03	188.2M	568.6K	1.0M	6.1K	178.3K

The portfolio has grown steadily across all metrics, with impressions and active content count increasing each month. AI sessions show the steepest growth trajectory — a trend that aligns with the broader industry shift toward AI-powered search interfaces.

What Weakened, Reversed, or Stayed Nuanced.

The paper keeps negative and mixed results visible because they improve decision quality and prevent overclaiming.

2 reversed myths	**2** nuanced myths	**1** direct debunk

These are not filler pages. They are the guardrails that keep the playbook honest. When a popular SEO belief did not survive contact with the portfolio, we kept the negative result instead of smoothing it into a more marketable story.

REVERSED **High search volume is not the safest demand target**
Lower-volume demand often converts into healthier pages because it is easier to win and match intent.

REVERSED **Optimization flags mark leverage, not failure**
Flagged content is often already visible enough to measure, which makes it a better optimization target.

NUANCED **Word count works as a threshold, not a guarantee**
Depth helps until the topic is sufficiently covered. After that, extra length produces diminishing returns.

DEBUNKED **AI-generated content is not penalized by default**
Quality and model/process differences matter more than whether an AI helped draft the page.

NUANCED **Freshness amplifies quality, it does not replace it**
Refreshing strong pages works far better than refreshing thin pages that were weak to begin with.

"High Search Volume = More Traffic."

Search volume is a weak expectation-setting number on its own. The local active-content sample shows that pages often outpace the nominal keyword-volume benchmark.

HEALTH SCORE BY SEARCH VOLUME BUCKET

SV BUCKET	COUNT	AVG SV	AVG IMP	IMP > SV	HEALTH	POS
1-100	25.4K	22	3.5K	87.88	44.4	13.4
100-1K	4.0K	302	3.7K	64.94	41.9	16.8
1K-10K	874	2.9K	3.2K	29.75	39.7	20.7
10K+	119	38.1K	6.9K	7.56	43.6	21.1

82.89%
pages above keyword volume

0.0083
raw SV ” imp correlation

-0.0419
log SV ” imp correlation

The local active-content sample outpaces the keyword-volume benchmark more often than not. Specifically, 82.89% of pages with non-zero volume show impressions above the stored search-volume figure.

The raw correlation between stored search volume and page impressions is 0.0083, and the log-scaled version is -0.0419: both are weak. Low-volume buckets still look easiest to outperform, while the highest-volume terms remain the least likely to beat their benchmark. That makes search volume more useful as a rough competition hint than as a literal traffic forecast.

> **Use search volume as a competition hint, not a literal traffic forecast**
> Why: Local page-level performance only weakly tracks the stored keyword-volume benchmark
> How: Start from beatable demand and clear intent match, then use search volume as a rough prior instead of treating it as the page's expected ceiling.
> **Expected: Improves the chance of targeting topics where real page-level upside is understated by keyword tools.**
> Measure: Track average position, page-level impressions, and the share of pages exceeding their nominal keyword-volume benchmark.

"Content With Flags Is Failing."

Internal optimization flags are workflow signals, not natural definitions of content quality.

HEALTH SCORE BY NUMBER OF ACTIVE OPTIMIZATION FLAGS

This was one of the most counterintuitive findings in the paper. Content with **2-3 optimization flags can score better** than content with zero flags.

The reason is structural: many flags can only trigger on pages that already have enough impressions or behavior data to diagnose a concrete issue. A page with zero visibility may receive no tactical flags at all.

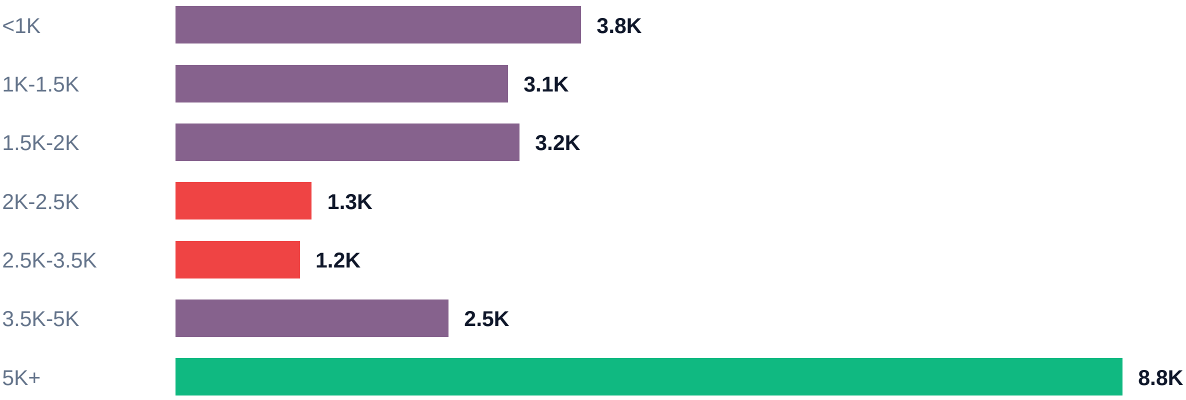
Implication: flagged content is often better optimization material than invisible content, but the signal should be read as workflow priority rather than a public claim that flagged pages are inherently stronger.

> **Treat measurable optimization issues as priority opportunities**
> Why: A page with a specific diagnosed issue is usually easier to improve than a page with no measurable demand
> How: Review visible pages for the exact problem they show, such as weak click capture or thin coverage, then fix that issue first instead of rewriting the whole asset.
> **Expected: Focuses effort on pages where the upside is knowable and more immediate.**
> Measure: Track impressions, clicks, and average position before and after each targeted fix.

"Longer Content Always Ranks Better."

The cleaner local pattern is visibility, not a universal rule. Longer content tends to earn more impressions and clicks, but that still does not make word count a guarantee.

AVERAGE IMPRESSIONS BY WORD COUNT BUCKET



WORDS	PAGES	AVG IMP	CLICKS	SESSIONS	AI PAGE %
<1K	12.3K	3.8K	12.88	14.58	5.47
1K-1.5K	7.2K	3.1K	10.66	14.67	3.1
1.5K-2K	7.0K	3.2K	11.15	15.8	4.39
2K-2.5K	1.8K	1.3K	5.98	5.95	2.39
2.5K-3.5K	14.5K	1.2K	5.91	7.98	2.8
3.5K-5K	9.8K	2.5K	9.08	19.56	7.1
5K+	9.4K	8.8K	26.9	94.02	25.11

The local active-content snapshot shows a **visibility gradient, not a magic threshold.**The strongest traffic bucket here is 5K+, which leads impressions, clicks, sessions, and AI-page incidence.

The takeaway stays nuanced: depth clearly correlates with visibility in this sample, but that does not mean every topic needs 5,000+ words. Quality depth still beats arbitrary length, and the local cached exports do not support a clean page-level duration sweet spot yet.

Use depth deliberately on pages that already show demand
Why: Longer buckets in the local active-content sample carry materially more impressions, clicks, and AI incidence
How: Expand thin but visible pages with missing subtopics, examples, comparisons, and evidence instead of padding every page to a fixed word target.
Expected: Improves visibility where additional depth is most likely to matter.
Measure: Track impressions, clicks, sessions, and AI page incidence by refreshed word-count bucket.

"AI-Generated Content Is Penalized."

Within this mostly AI-authored portfolio, age-controlled model cohorts do not show a simple blanket penalty tied only to AI use.

This is perhaps the most debated topic in SEO circles: does Google penalize AI-generated content? Our dataset of 341,018 pieces — almost all AI-generated across 5 different models — provides a unique perspective.

When we compared models inside the same age tiers, the differences moved around rather than pointing to a single winner. Both stronger and weaker outcomes show up inside AI-authored cohorts. That means the meaningful distinction is process quality, not a simple AI-versus-human split.

This portfolio does not show a blanket penalty tied only to AI use. The safer interpretation is that model choice, editing standards, and topic fit drive the differences we see. Quality controls matter far more than whether a given draft came from one AI workflow or another.

> **Evaluate publishing systems by results, not ideology**
> Why: The differences in this portfolio come from model and process quality, not a simple yes-or-no AI penalty
> How: Test writing systems within the same topic and publication window, then compare how they perform after launch instead of assuming one method is inherently safer.
> **Expected: Keeps content operations evidence-led and avoids false binary decisions about AI use.**
> Measure: Track impressions, average position, and quality-review outcomes by workflow cohort.

"Keyword Difficulty Is a Reliable Predictor."

Competition level matters, but the difference is smaller than you'd expect — and the growth ratio tells the real story.

COMPETITION	HEALTH	AVG IMP	AVG POS	GROWING	DECLINING	COUNT
LOW	23.8	1.8K	15.7	58.3K	27.2K	226.0K
HIGH	22.9	976	18.5	6.7K	5.3K	27.0K
MEDIUM	22.6	1.2K	20.2	5.3K	3.0K	19.9K

LOW competition: health 23.8, growth-to-decline ratio 2.1:1. HIGH competition: health 22.9, ratio 1.3:1. The health difference is small (only 0.9 points), but the **growth ratio tells the real story**.

LOW competition keywords are 62% more likely to grow than HIGH competition ones. The competition metric works — but not as a ranking predictor. It's a **growth probability indicator**.

"Fresh Content Always Outperforms."

Freshness amplifies quality — it doesn't replace it. Thin content stays thin regardless of update date.

FRESHNESS	WORD COUNT	COUNT	HEALTH	AVG IMP	POS
0-30	1000-2000	25.6K	34.1	1.9K	16.6
0-30	2000-3500	32.3K	26.3	691	12.5
0-30	3500+	90.1K	28.7	3.5K	17.2
0-30	<1000	151	14.6	66	38.1
31-90	1000-2000	9.3K	12.5	51	32.2
31-90	2000-3500	24.2K	25.3	522	11.9
31-90	3500+	16.9K	24.1	1.2K	16.6
31-90	<1000	415	13.5	164	52
91-180	1000-2000	6.8K	9.9	2	10.8
91-180	2000-3500	2.7K	15.9	690	13
91-180	3500+	19.7K	21.8	963	21.4
91-180	<1000	26.0K	20.3	6	9.1
181+	1000-2000	41.3K	9.3	13	19.2
181+	2000-3500	1.3K	12.6	62	17.7
181+	3500+	41.0K	15.9	684	14.7
181+	<1000	4.0K	8.2	1	14.2

When we controlled for word count, the picture changed materially. At 3500+ words, fresh content (0-30d) scores 28.7 health vs stale (181+) at 15.9 — a 12.8-point gap. At <1000 words, fresh scores 14.6vs stale at 8.2 — a 6.4-point gap, but that thin-content bucket is very small.

Freshness multiplies existing quality. Refreshing a comprehensive, well-structured piece produces dramatic results. Thin pages can still improve when refreshed, but the upside is lower and the evidence base is much smaller. This means refresh efforts should be prioritized on your best content first.

What Actually Correlates?

Exploratory appendix: Pearson correlation matrix across 16 numeric features from 61.8K content pieces with active sessions and impressions. Direct aggregate comparisons elsewhere in the paper remain the primary evidence base.

FEATURE CORRELATION HEATMAP (PEARSON R)

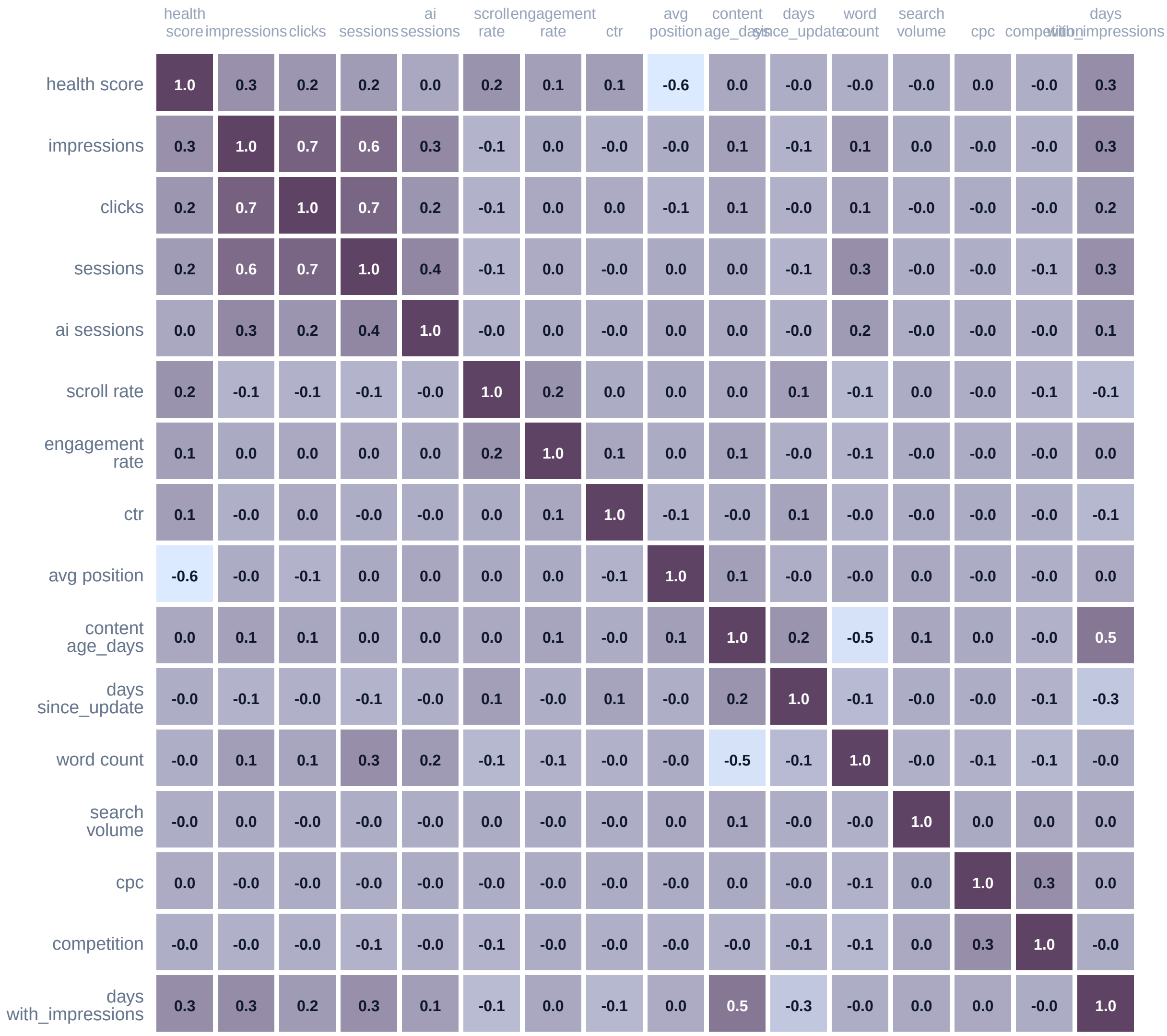


Chart read: Dark cells mark features that tend to move together inside the active-content sample. Use this to spot overlapping signals and avoid treating highly related metrics as separate independent wins.

STRONGEST POSITIVE CORRELATIONS

Impressions × Clicks: **r=0.708**
Clicks × Sessions: **r=0.666**
Impressions × Sessions: **r=0.62**
Content Age × Days Visible: **r=0.496**
Sessions × ai Sessions: **r=0.354**

STRONGEST NEGATIVE CORRELATIONS

health score × Average Position: **r=-0.592**
Content Age × Word Count: **r=-0.517**
Days Since Update × Days Visible: **r=-0.26**
Scroll Depth × Days Visible: **r=-0.141**
Days Since Update × Word Count: **r=-0.132**

Chart read: The lists below summarize the strongest pairings from the heatmap. Treat them as descriptive relationships inside the sample, not as proof that changing one metric will automatically move the other.

The Content Archetypes.

Exploratory k-means clustering (k=5) groups the sampled active-content set into 5 segments. These are descriptive clusters, not production-ready personas.

Cluster 1 (3% of content, n=1.9K)

Exploratory segment with 3% of the sampled ML set (n=1856).

Original heuristic label in the analysis script: Champions

Health: **53.09** Impressions: **37.9K** Position: **16.4** Age: **196d** Words: **5.0K**

Cluster 2 (14.9% of content, n=9.2K)

Exploratory segment with 14.9% of the sampled ML set (n=9211).

Original heuristic label in the analysis script: Rising Stars

Health: **48** Impressions: **928** Position: **12.6** Age: **180d** Words: **2.3K**

Cluster 3 (24% of content, n=14.9K)

Exploratory segment with 24% of the sampled ML set (n=14857).

Original heuristic label in the analysis script: Rising Stars

Health: **47** Impressions: **3.9K** Position: **14.7** Age: **313d** Words: **689**

Cluster 4 (10.8% of content, n=6.7K)

Exploratory segment with 10.8% of the sampled ML set (n=6657).

Original heuristic label in the analysis script: Rising Stars

Health: **43.2** Impressions: **2.0K** Position: **12.7** Age: **137d** Words: **2.3K**

Cluster 5 (47.3% of content, n=29.2K)

Exploratory segment with 47.3% of the sampled ML set (n=29209).

Original heuristic label in the analysis script: Rising Stars

Health: **39.9** Impressions: **2.3K** Position: **15.2** Age: **97d** Words: **3.8K**

Chart read: Each card is one cluster profile from the sampled active-content set. Compare the mixes of visibility, age, and depth across cards to understand common content states, but do not treat these groups as fixed audience personas.

These clusters are useful for exploratory segmentation, but the names in the original analysis were heuristic and sometimes duplicated. For public reporting, the safer use is to compare cluster profiles, not to treat them as fixed operational archetypes.

What Predicts Health?

Random Forest feature importance for predicting health score. The model is holdout-tested, but the target itself is partly constructed from some of these inputs, so importance is descriptive rather than causal.

FEATURE IMPORTANCE (RANDOM FOREST ’ HEALTH SCORE)

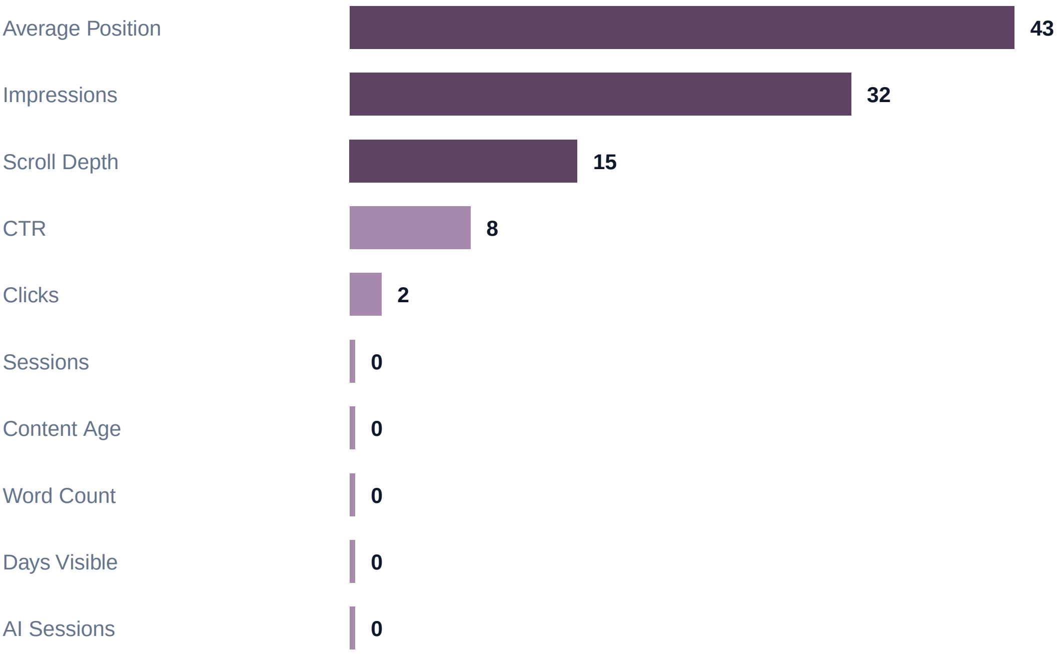


Chart read: Higher bars mean the model leaned more heavily on that feature when estimating health score. Because health score already includes some visibility inputs, read this as model behavior, not as a standalone optimization order.

Average Position is the #1 predictor of health score at 43% importance, followed by Impressions (32%) and Scroll Depth (15%). This ranking shows which features the model uses most, though note that health score is partly constructed from inputs such as position and impressions. High importance is therefore expected and does not imply external causation.

PCA 2D PROJECTION — COLORED BY CLUSTER

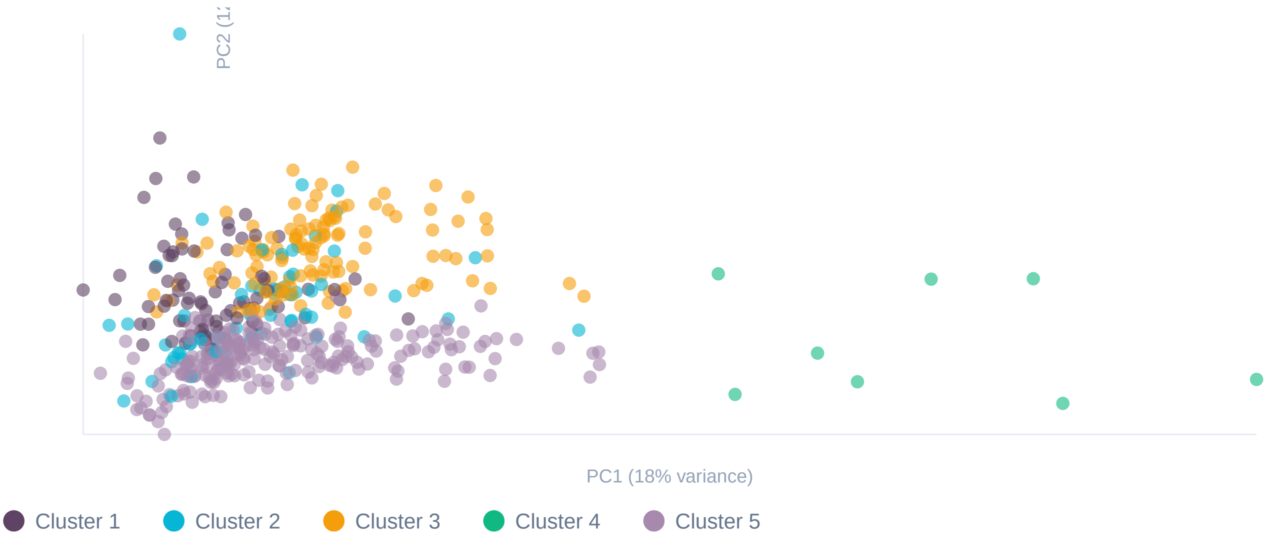


Chart read: Each dot is one sampled page projected into two simplified dimensions. Tighter color groupings mean the cluster is more internally similar, while overlap means the segment boundaries are fuzzy rather than cleanly separated.

What Predicts Growth?

Logistic regression (71% holdout accuracy) describing which sampled features separate growing from declining pages.

GROWTH PREDICTION COEFFICIENTS (LOGISTIC REGRESSION)

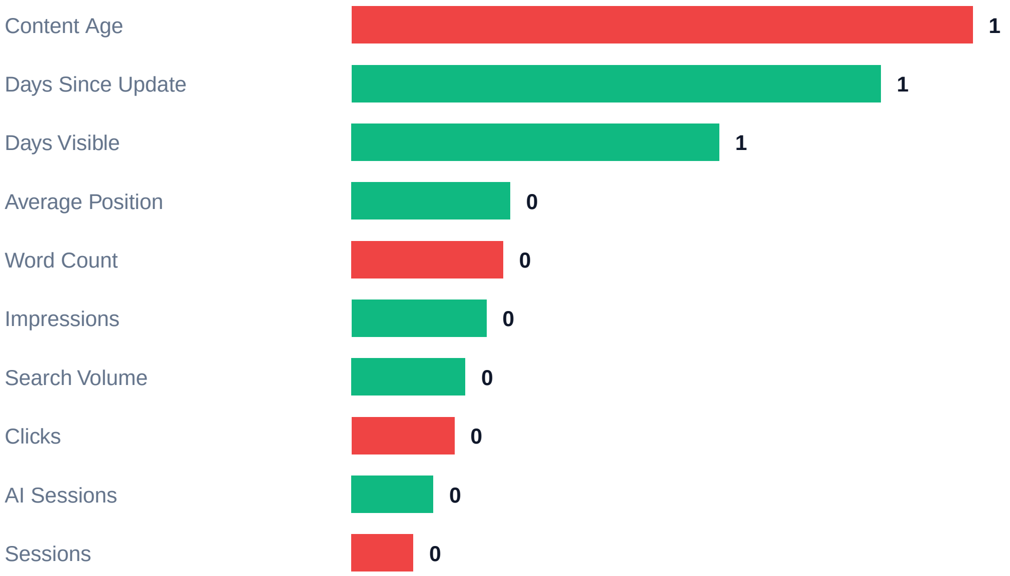


Chart read: Green bars point toward growth in this model and red bars point away from it. The bar size shows relative influence inside the sampled dataset, so use it to prioritize investigation rather than to assume a guaranteed lever.

Content age is the strongest negative signal in this model, while days visible and recent impressions are among the strongest positive signals. Read these as descriptive indicators from the sampled active-content set, not as direct instructions to optimize one variable in isolation.

The shallow decision tree is now holdout-tested as well, but it should still be treated as a simple descriptive ruleset. It is most useful for seeing which combinations of visibility, clicks, and engagement separate healthier pages from weaker ones in the sample.

TOP 10% VS BOTTOM 10% CONTENT PROFILE

DECILE	HEALTH	AVG IMP	POS	AGE	FRESH	WORDS	COUNT
bottom_10pct	5	0	-	97d	81d	2.6K	34.2K
top_10pct	57.1	10.3K	6.3	201d	55d	3.2K	34.2K

Chart read: This comparison shows what the strongest decile looks like versus the weakest one. Use it as a benchmark gap analysis: if your page profile resembles the lower row, you know which broad traits still need work.

The contrast is useful but still descriptive: stronger pages in this sample tend to be fresher, longer, and much better positioned than weaker ones. The table is best used as a benchmark profile, not as a guaranteed formula.

Metric **Distributions.**

Normalized percentile view of the sampled active-content set. Each row is scaled independently so smaller metrics remain readable.

PERCENTILE BOX PLOTS (P10 / P25 / P50 / P75 / P90)

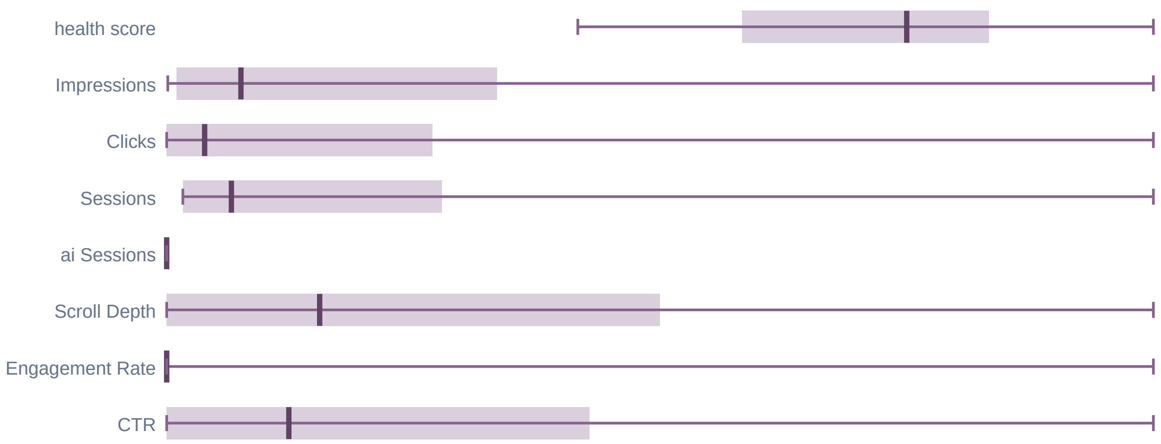


Chart read: Each row is scaled independently, so the shape matters more than the raw bar length. Use the median and upper-percentile markers to judge whether a page sits near the middle of the pack or in the stronger tail of the portfolio.

METRIC	P10	P25	MEDIAN	P75	P90	MEAN
health score	25	35	45	50	60	43.57
Impressions	10	91	668	2972	8878.1	3510.32
Clicks	0	0	1	7	26	12.12
Sessions	1	1	4	17	61	25.75
ai Sessions	0	0	0	0	0	0.26
Scroll Depth	0	0	8.86	28.57	57.14	20.28
Engagement Rate	0	0	0	0	8.7	3.34
CTR	0	0	0.13	0.45	1.05	0.7

Use these percentiles as directional benchmarks for the active-content sample rather than universal thresholds. The chart is normalized for readability; the table holds the actual numeric values.

Priority Actions.

Ranked by the clearest, most defensible patterns in this study. Start with the actions that preserve or extend existing visibility.

1. **Refresh mature pages before they decay**
   The refresh effect is one of the clearest signals in the dataset: older pages that were updated recently outperform similarly old stale pages by a wide margin.

2. **Improve page-one click capture before rebuilding from scratch**
   Pages that already rank well can often return clicks faster through snippet, clarity, and relevance improvements than net-new content can.

3. **Prioritize lower-competition commercial intent where possible**
   Commercial and transactional low-competition topics outperform more crowded configurations on both health context and impressions.

4. **Use content depth where the topic requires it**
   The strongest lift happens when thin but visible pages become genuinely useful, not when pages are padded for length alone.

5. **Distribute proven pages across more than one channel**
   The strongest pages in this portfolio rarely rely on search alone. Support strong pages after publication.

6. **Use optimization flags as workflow cues, not quality judgments**
   Measured issues on already-visible pages are usually more actionable than pages with no measurable demand.

7. **Monitor AI-referral content as a separate visibility layer**
   AI-attracting pages behave differently from classic organic winners, so they deserve their own monitoring and editorial review.

8. **Resolve cannibalization where multiple pages split the same demand**
   73.2M impressions are tied up in overlapping demand. Consolidation and clearer targeting can release that value.

The Content Refresh Playbook.

Step-by-step process backed by our freshness and age analysis findings.

1. Identify refresh candidates

Why: Content declines from 33.1 at 61-90 days to 14 by 271-365 days in the local age curve
How: List older pages that used to earn impressions or still have meaningful visibility. Start with pages that are clearly dated, incomplete, or no longer competitive.
Expected: Targets the highest-value content most likely to recover
Measure: Track refreshed-page count, health score lift, and 30-day impression recovery versus untreated candidates.

2. Audit and expand thin sections

Why: Top 10% content averages 3.2K words versus 2.6K in the bottom 10%
How: Read the page end to end, identify weak sections, then add missing definitions, examples, comparisons, and current evidence instead of padding.
Expected: Improves depth where missing coverage is holding back an already-visible page
Measure: Track word-count expansion, health score change, and impression lift by refresh batch.

3. Update all statistics, dates, and references

Why: The strongest measured freshness window is 31-90 days at 7.88:1 growth to decline
How: Replace stale figures, update named sources, add recent examples, fix broken links, and make the page visibly current for both users and search engines.
Expected: Aligns the page with the strongest measured freshness band in this dataset
Measure: Track time-to-reindex, 30-day impression lift, and CTR lift after the update goes live.

4. Improve engagement elements

Why: High scroll + high engagement outperforms the weakest bucket by 11.2 health points
How: Add table-of-contents links where useful, break dense sections into scannable blocks, and use visuals or callouts where they actually help comprehension.
Expected: Supports the engagement patterns associated with stronger visibility
Measure: Track scroll depth, engagement rate, and health score before and after layout changes.

5. Re-submit and monitor

Why: Updated content still needs re-crawl and a monitored follow-up window
How: Request recrawl where appropriate, republish the update through your normal distribution channels, and check the page again at 30, 60, and 90 days.
Expected: Keeps the refresh program tied to measured follow-up windows instead of one-time edits
Measure: Track reindex confirmations, ranking recovery, and 30/60/90-day health score trend.

AI Traffic Optimization Guide.

Treat AI referrals as a real but still small traffic layer, and optimize around what is measured rather than around assumed platform rules.

In our dataset, content attracting AI sessions has a distinct profile: higher average impressions, deeper ranking positions, and stronger portfolio-level visibility than content without AI referrals. The scale caveat matters: AI referrals account for only 1.06% of tracked sessions in the local portfolio snapshot.

What Our Data Shows About AI-Attracting Content

High-AI content in our portfolio averages a weaker Google position but much higher impressions than no-AI content. Inside the local active-content sample, the highest AI page rates appear in commercial intent and the 5K+ word-count bucket.

Practical Optimization Steps

1. Monitor AI referrals separately from organic clicks and pageview totals.
2. Prioritize clear factual statements, named evidence, and easy-to-scan structure.
3. Review provider mix regularly: OpenAI 4.2K | Gemini 2.3K | Perplexity 781 | Copilot 117 | Claude 83.
4. Focus first on reference-style pages that already carry broad visibility.
5. Treat monetization claims cautiously unless attribution is session-level and clean.

Important caveat: we do not treat revenue as a headline proof point for AI traffic here. The available export is a content-level bucket, not a clean session-level attribution model, so the reliable use case is visibility analysis rather than direct monetization forecasting.

The Quick Wins **Checklist.**

Actions you can take this week based on our findings.

- **TODAY** — **Review measured optimization issues on already-visible pages**
  Pages with specific, observable issues are often the fastest wins because the upside is measurable.
- **TODAY** — **Review your best 20 striking-distance pages**
  Pages close to page 1 are usually better candidates for focused improvement than low-visibility pages with no traction.
- **THIS WEEK** — **Refresh your oldest high-visibility pages**
  Older pages with prior visibility are the cleanest candidates for a recency and depth update.
- **THIS WEEK** — **Improve readability on top-performing long pages**
  Clear navigation, sectioning, and scanability support the engagement patterns associated with stronger pages.
- **THIS MONTH** — **Consolidate overlapping pages that split the same demand**
  Choose the strongest page, merge overlapping coverage, and remove avoidable competition between your own URLs.
- **THIS MONTH** — **Expand high-potential thin pages**
  Focus on the thin pages that already earn impressions instead of trying to bulk-expand low-demand inventory.
- **ONGOING** — **Keep a rolling refresh calendar**
  The strongest freshness effects appear when updates happen before a page has fully decayed.
- **ONGOING** — **Support new and strong pages with distribution**
  Email, social, internal links, and referral placement help strong pages become durable assets instead of one-channel pages.

Structural **Risks.**

These are the portfolio-wide issues most likely to suppress performance even when individual pages are strong.

703.2K cannibalizing queries	**73.2M** impressions at risk	**274.4K** clicks at risk

Chart read: This bar shows how the cannibalization cases are distributed by severity. A larger red share means more overlapping demand needs consolidation or clearer page targeting first.

INDEXING COVERAGE

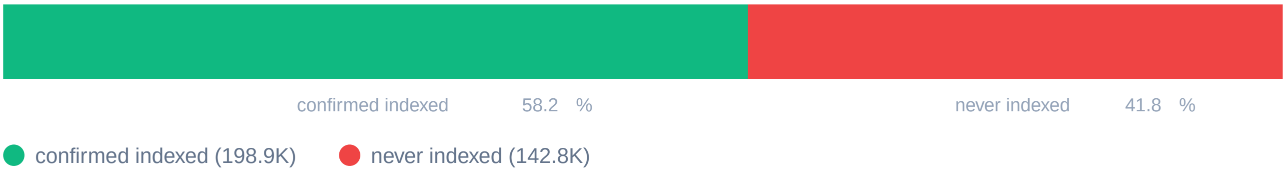


Chart read: This bar shows how much of the portfolio is actually available to compete in search. If the indexed share is smaller than expected, fixing crawl and indexing issues can unlock results before new content creation does.

Methodology & Limitations.

Data Sources

Google Search Console (GSC) and Google Analytics 4 (GA4) data aggregated in BigQuery through a read-only MCP service account with SELECT-only guardrails.

Study Scope

90-day performance window (rolling), 30-day trend comparison, local historical monthly series, and local active-content feature-vector cuts where noted.

Evidence Standard

Headline findings prioritize direct aggregate comparisons. ML pages are exploratory appendix material and do not override direct portfolio evidence.

Health Score

Impressions (30 pts) + position (30 pts) + CTR (20 pts) + scroll depth (20 pts).

ML Pipeline

61.8K content pieces with sessions > 0 and impressions > 0 in the local feature-vector snapshot. Sklearn: K-Means (k=5), Random Forest (80/20 split), Logistic Regression (80/20 split), PCA (2 components), and a shallow Decision Tree (80/20 split). ML pages remain exploratory and secondary to direct aggregate comparisons.

Statistical Approach

Pearson correlation for numeric pairs. Minimum n=50 per analysis bucket (n=30 for cross-correlations). APPROX_QUANTILES for percentiles. Weighted CTR is used where total clicks and impressions are available. No p-values or confidence intervals are reported.

Confounding Variables

Content age confounds model-performance comparisons. Revenue tracking covers 8 of 57 clients, so revenue is treated cautiously. Known-referral rules may miss some AI sources, and several extended cuts come from the local active-content feature-vector subset rather than full inventory.

Limitations

Observational study: correlations do not prove causation. Health score is a FlyRank composite metric, not a Google-endorsed standard. The paper uses the local cached snapshot rather than fresh warehouse pulls. The cached feature vector does not include page-level engagement seconds or a full ranking-query universe, so duration and primary-keyword relevance sections are not claimed here. Legacy exports were excluded or demoted when provenance or metric meaning was not strong enough for public use, including the backlink page in this version.